MIJ TECHNOLOGIES • PYTHON PROGRAMMING SERIES

ASSIGNMENT 5

Hospital Patient Billing System

TOTAL MARKS	DIFFICULTY	ESTIMATED TIME	TOPICS COVERED
36 Marks	Very Hard	6 - 8 hours	Functions, Loops & Conditionals

DESIGN REQUIREMENTS: Your solution must demonstrate structured programming with proper use of functions, loops, and conditionals. Clearly separate responsibilities into different functions — avoid writing everything in one long block of code. Validate all input (including insurance status against provider) and use meaningful variable names. Your program must run without crashing.

Overview

Write a Command Line Program that simulates a Hospital Patient Billing System. A receptionist enters one or more patient visits. For each visit the program calculates the service charge, adds the optional lab and medication charges, applies the insurance coverage, prints the bill, and after the last patient prints a daily hospital report. All rules are in the tables below; every amount is in UGX.

A PART A — Patient Information Input — 6 Marks

For every patient the program asks for the fields below, **in this order**. On an invalid entry it prints **Invalid input. Please try again.** and asks for the **same** field again. The program always collects at least one patient, then asks **Add another patient? (yes/no)** after each bill and stops on **no**.

Table A1 — Validation.

#	Prompt (exact text)	Accepted values	Case and spaces
1	Enter Patient Name:	At least 1 non-space character	Spaces trimmed
2	Enter Service Type (Consultation/Emergency/Surgery/Admission):	Consultation, Emergency, Surgery, Admission	Not case-sensitive
3	Enter Hours: (Consultation, Emergency, Surgery) or Enter Days: (Admission)	Whole number from 1 to 24 hours, or from 1 to 30 days	Spaces trimmed
4	Insurance Status (Yes/No):	yes or no	Not case-sensitive
5	Insurance Provider (NHIF/AAR/Jubilee/None):	Status Yes: NHIF, AAR or Jubilee. Status No: None	Not case-sensitive
6	Lab Tests? (Yes/No):	yes or no	Not case-sensitive
7	Medication Pack? (Yes/No):	yes or no	Not case-sensitive
8	Add another patient? (yes/no):	yes or no	Not case-sensitive

Table A2 — Standard input rules and glossary.

#	Rule or term	Meaning
S1	Whole number	Digits only. Decimals, letters, negative numbers, 0 and empty input are invalid.
S2	Status and provider	Status Yes with provider None is invalid. Status No with NHIF, AAR or Jubilee is invalid. Only the provider is asked again.
S3	TOTAL BILL	Service charge plus the lab and medication charges (Parts B and D).
S4	Patient pays	TOTAL BILL minus insurance coverage. This is the hospital's revenue from the patient.
S5	Robustness	The program never crashes and never skips a field.

B PART B — Service Pricing 6 Marks

Table B1 — Service rates.

Service	Hours or days in the band	Rate
Consultation	Hour 1	50,000 UGX per hour
	Hour 2 and above	30,000 UGX per hour
Emergency	Hours 1 - 2	100,000 UGX per hour
	Hour 3 and above	80,000 UGX per hour
Surgery (flat)	Minor: 1 - 2 hours	500,000 UGX for the whole surgery
	Major: 3 hours and above	1,200,000 UGX for the whole surgery
Admission	Days 1 - 3	150,000 UGX per day
	Days 4 - 7	120,000 UGX per day
	Day 8 and above	100,000 UGX per day

Applying the table. Consultation, Emergency and Admission use **graduated** pricing: every hour or day is charged the rate of the band its own number falls in (a 10-day admission is 3 x 150,000 + 4 x 120,000 + 3 x 100,000). Surgery uses **whole-quantity** pricing: the duration selects one band and the flat rate of that band is the whole service charge.

C PART C — Insurance Coverage Calculation 6 Marks

Table C1 — Providers. Coverage is worked out on the **total bill** (service charge plus extras).

Provider	Coverage	Maximum cap
NHIF	60% of the total bill	300,000 UGX
AAR	80% of the total bill	No limit
Jubilee	70% of the total bill	500,000 UGX
None	0% (the patient pays 100%)	Not applicable

```
Insurance Coverage = smaller of ( TOTAL BILL x Coverage % , Maximum Cap )
Patient Pays       = TOTAL BILL - Insurance Coverage
```

Every total bill is a multiple of 10,000 UGX, so every coverage amount is a whole number and no rounding is needed.

D PART D — Optional Lab and Medication Charges 6 Marks

Table D1 — Extras. Each extra is a flat charge for the whole visit, added to the service charge **before** insurance is applied, so insurance also covers the extras.

Extra	Asked as	Charge
Lab Tests	Lab Tests? (Yes/No)	40,000 UGX
Medication Pack	Medication Pack? (Yes/No)	30,000 UGX

```
TOTAL BILL = Service Charge + Lab Tests (if yes) + Medication Pack (if yes)
```

E PART E — Patient Bill Display 6 Marks

For every patient print the three blocks shown in the sample output (service charge breakdown, insurance coverage, patient bill). Use the labels in Table E1 exactly.

Printed label	Value shown	Rule
Patient / Service / Duration	As entered	Surgery is shown as Surgery (Minor) or Surgery (Major)
Service Charge	Amount UGX	Table B1
Lab Tests / Medication Pack	Amount UGX	Printed only when the answer was yes
TOTAL BILL	Amount UGX	Service charge plus extras
Insurance Provider	Name	NHIF, AAR, Jubilee or None
Insurance Coverage	Amount UGX	Part C, 0 UGX when the provider is None
Patient Pays	Amount UGX	TOTAL BILL minus coverage

F PART F — Daily Hospital Report 6 Marks

After the answer **no**, print the report in Table F1.

Printed label	What is counted	Format
Total Patients Served	Every patient entered	Whole number
Total Revenue Collected	Sum of Patient Pays of all patients	Whole UGX
Average Bill Per Patient	Total revenue / patients, rounded half up	2 decimals, UGX
Total Insurance Coverage	Sum of Insurance Coverage of all patients	Whole UGX
Service Breakdown	Patients per service, table order, 0 included	Name: n | Name: n
Most Common Service	Service with the most patients	Name
With / Without Insurance	Patients with status Yes / No	Whole numbers
Most Common Provider	Provider with the most insured patients (NHIF, AAR, Jubilee only)	Name

Table F2 — Edge cases.

Situation	Exact output
Two or more services (or providers) have the same highest count	All names in table order joined by " / ", then " (tie)"
No patient has insurance	Most Common Provider prints **None**
A service has 0 patients	It is still printed, with 0
Zero patients entered	Cannot happen: at least one patient is always collected

Test Cases For Verification

EXAMPLE Case 1 — Consultation, 1 hour, No Insurance

```
Hour 1          : 1 x 50,000 = 50,000 UGX
TOTAL BILL      : 50,000 UGX
Patient Pays    : 50,000 - 0 = 50,000 UGX
```

EXAMPLE Case 2 — Admission, 10 days, NHIF

```
Days 1-3        : 3 x 150,000 = 450,000 UGX
Days 4-7        : 4 x 120,000 = 480,000 UGX
Days 8-10       : 3 x 100,000 = 300,000 UGX
TOTAL BILL      : 1,230,000 UGX
NHIF (60%)      : 1,230,000 x 0.60 = 738,000 UGX -> CAPPED at 300,000
Patient Pays    : 1,230,000 - 300,000 = 930,000 UGX
```

EXAMPLE Case 3 — Surgery, 4 hours, AAR

```
Surgery (Major) : 1,200,000 UGX
TOTAL BILL      : 1,200,000 UGX
AAR (80%)       : 1,200,000 x 0.80 = 960,000 UGX
Patient Pays    : 1,200,000 - 960,000 = 240,000 UGX
```

EXAMPLE Case 4 — Emergency, 5 hours, NHIF

```
Hours 1-2       : 2 x 100,000 = 200,000 UGX
Hours 3-5       : 3 x 80,000 = 240,000 UGX
Lab Tests       : 40,000 UGX
Medication Pack : 30,000 UGX
TOTAL BILL      : 510,000 UGX
NHIF (60%)      : 510,000 x 0.60 = 306,000 UGX -> CAPPED at 300,000
Patient Pays    : 510,000 - 300,000 = 210,000 UGX
```

EXAMPLE Case 5 — Surgery, 4 hours, Jubilee

```
Surgery (Major) : 1,200,000 UGX
TOTAL BILL      : 1,200,000 UGX
Jubilee (70%)   : 1,200,000 x 0.70 = 840,000 UGX -> CAPPED at 500,000
Patient Pays    : 1,200,000 - 500,000 = 700,000 UGX
```

Table T1 — Duration boundaries (no insurance, no extras: Patient Pays = TOTAL BILL).

Service	Duration	Service charge	Service	Duration	Service charge
Consultation	1 hour	50,000 UGX	Surgery	3 hours	1,200,000 UGX
Consultation	2 hours	80,000 UGX	Surgery	24 hours	1,200,000 UGX
Consultation	24 hours	740,000 UGX	Admission	1 day	150,000 UGX
Emergency	1 hour	100,000 UGX	Admission	3 days	450,000 UGX
Emergency	2 hours	200,000 UGX	Admission	4 days	570,000 UGX
Emergency	3 hours	280,000 UGX	Admission	7 days	930,000 UGX
Emergency	24 hours	1,960,000 UGX	Admission	8 days	1,030,000 UGX
Surgery	2 hours	500,000 UGX	Admission	30 days	3,230,000 UGX

Table T2 — Insurance coverage by total bill (UGX covered).

TOTAL BILL	NHIF 60%, cap 300,000	AAR 80%, no cap	Jubilee 70%, cap 500,000	None
50,000	30,000	40,000	35,000	0
500,000	300,000	400,000	350,000	0
510,000	300,000	408,000	357,000	0
1,200,000	300,000	960,000	500,000	0

EXAMPLE **Report of the 5 patients in the sample run**

```
Total Revenue  = 22,000 + 500,000 + 970,000 + 700,000 + 10,000 = 2,202,000 UGX
Total Coverage = 88,000 + 0 + 300,000 + 500,000 + 40,000 = 928,000 UGX
Average Bill   = 2,202,000 / 5 = 440,400.00 UGX
```

Sample Terminal Output

Print your output with full-width separators and fixed-width columns, exactly as shown. Each card is one patient and shows re-prompts; the last card is the daily report.

TERMINAL — PART 1 OF 6 (Consultation, AAR)

```
================================================================================
Enter Patient Name: Alice Nambi
Enter Service Type (Consultation/Emergency/Surgery/Admission): Dental
Invalid input. Please try again.
Enter Service Type (Consultation/Emergency/Surgery/Admission): Consultation
Enter Hours: 3
Insurance Status (Yes/No): maybe
Invalid input. Please try again.
Insurance Status (Yes/No): Yes
Insurance Provider (NHIF/AAR/Jubilee/None): AAR
Lab Tests? (Yes/No): No
Medication Pack? (Yes/No): No

================================================================================
   SERVICE CHARGE BREAKDOWN
================================================================================
  Hour 1                  : 1 x 50,000          =   50,000 UGX
  Hours 2-3               : 2 x 30,000          =   60,000 UGX
  TOTAL BILL              = 110,000 UGX

================================================================================
   INSURANCE COVERAGE
================================================================================
  Provider                : AAR (80%, no cap)
  Calculated              : 110,000 x 80% = 88,000 UGX
  Insurance Coverage      : 88,000 UGX
  Patient Pays            = 110,000 - 88,000 = 22,000 UGX

================================================================================
   PATIENT BILL
================================================================================
  Patient                 : Alice Nambi
  Service                 : Consultation
  Duration                : 3 hours
  Service Charge          : 110,000 UGX
  TOTAL BILL              : 110,000 UGX
  Insurance Provider      : AAR
  Insurance Coverage      : 88,000 UGX
  Patient Pays            = 22,000 UGX
================================================================================
Add another patient? (yes/no): yes
```

TERMINAL — PART 2 OF 6 (Surgery, No Insurance)

```
================================================================================
Enter Patient Name: John Okello
Enter Service Type (Consultation/Emergency/Surgery/Admission): Surgery
Enter Hours: 2
Insurance Status (Yes/No): No
Insurance Provider (NHIF/AAR/Jubilee/None): NHIF
Invalid input. Please try again.
Insurance Provider (NHIF/AAR/Jubilee/None): None
Lab Tests? (Yes/No): No
Medication Pack? (Yes/No): No

================================================================================
   SERVICE CHARGE BREAKDOWN
================================================================================
  Minor Surgery (< 3 hrs)  : 500,000 UGX
  TOTAL BILL               = 500,000 UGX

================================================================================
   INSURANCE COVERAGE
================================================================================
  Provider                 : None (0%)
  Insurance Coverage       : 0 UGX
  Patient Pays             = 500,000 - 0 = 500,000 UGX

================================================================================
   PATIENT BILL
================================================================================
  Patient                  : John Okello
  Service                  : Surgery (Minor)
  Duration                 : 2 hours
  Service Charge           : 500,000 UGX
  TOTAL BILL               : 500,000 UGX
  Insurance Provider       : None
  Insurance Coverage       : 0 UGX
  Patient Pays             = 500,000 UGX
================================================================================
Add another patient? (yes/no): yes
```

TERMINAL — PART 3 OF 6 (Admission, NHIF)

```
================================================================================
Enter Patient Name: Peter Ssali
Enter Service Type (Consultation/Emergency/Surgery/Admission): Admission
Enter Days: 0
Invalid input. Please try again.
Enter Days: 10
Insurance Status (Yes/No): Yes
Insurance Provider (NHIF/AAR/Jubilee/None): None
Invalid input. Please try again.
Insurance Provider (NHIF/AAR/Jubilee/None): NHIF
Lab Tests? (Yes/No): x
Invalid input. Please try again.
Lab Tests? (Yes/No): Yes
Medication Pack? (Yes/No): No

================================================================================
   SERVICE CHARGE BREAKDOWN
================================================================================
  Days 1-3                : 3 x 150,000          =  450,000 UGX
  Days 4-7                : 4 x 120,000          =  480,000 UGX
  Days 8-10               : 3 x 100,000          =  300,000 UGX
  Lab Tests               : 40,000 UGX
  TOTAL BILL              = 1,270,000 UGX

================================================================================
   INSURANCE COVERAGE
================================================================================
  Provider                : NHIF (60%, cap 300,000 UGX)
  Calculated              : 1,270,000 x 60% = 762,000 UGX
  Cap Applied             : 300,000 UGX
  Insurance Coverage      : 300,000 UGX
  Patient Pays            = 1,270,000 - 300,000 = 970,000 UGX

================================================================================
   PATIENT BILL
================================================================================
  Patient                 : Peter Ssali
  Service                 : Admission
  Duration                : 10 days
  Service Charge          : 1,230,000 UGX
  Lab Tests               : 40,000 UGX
  TOTAL BILL              : 1,270,000 UGX
  Insurance Provider      : NHIF
  Insurance Coverage      : 300,000 UGX
  Patient Pays            = 970,000 UGX
================================================================================
Add another patient? (yes/no): yes
```

TERMINAL — PART 4 OF 6 (Surgery, Jubilee)

```
================================================================================
Enter Patient Name:
Invalid input. Please try again.
Enter Patient Name: Grace Akello
Enter Service Type (Consultation/Emergency/Surgery/Admission): Surgery
Enter Hours: 4
Insurance Status (Yes/No): Yes
Insurance Provider (NHIF/AAR/Jubilee/None): Jubilee
Lab Tests? (Yes/No): No
Medication Pack? (Yes/No): 5
Invalid input. Please try again.
Medication Pack? (Yes/No): No

================================================================================
   SERVICE CHARGE BREAKDOWN
================================================================================
  Major Surgery (>= 3 hrs) : 1,200,000 UGX
  TOTAL BILL               = 1,200,000 UGX

================================================================================
   INSURANCE COVERAGE
================================================================================
  Provider                 : Jubilee (70%, cap 500,000 UGX)
  Calculated               : 1,200,000 x 70% = 840,000 UGX
  Cap Applied              : 500,000 UGX
  Insurance Coverage       : 500,000 UGX
  Patient Pays             = 1,200,000 - 500,000 = 700,000 UGX

================================================================================
   PATIENT BILL
================================================================================
  Patient                  : Grace Akello
  Service                  : Surgery (Major)
  Duration                 : 4 hours
  Service Charge           : 1,200,000 UGX
  TOTAL BILL               : 1,200,000 UGX
  Insurance Provider       : Jubilee
  Insurance Coverage       : 500,000 UGX
  Patient Pays             = 700,000 UGX
================================================================================
Add another patient? (yes/no): maybe
Invalid input. Please try again.
Add another patient? (yes/no): yes
```

TERMINAL — PART 5 OF 6 (Consultation, AAR)

```
================================================================================
Enter Patient Name: David Mugisha
Enter Service Type (Consultation/Emergency/Surgery/Admission): Consultation
Enter Hours: abc
Invalid input. Please try again.
Enter Hours: 1
Insurance Status (Yes/No): Yes
Insurance Provider (NHIF/AAR/Jubilee/None): AAR
Lab Tests? (Yes/No): No
Medication Pack? (Yes/No): No

================================================================================
   SERVICE CHARGE BREAKDOWN
================================================================================
  Hour 1                  : 1 x 50,000          =   50,000 UGX
  TOTAL BILL              = 50,000 UGX

================================================================================
   INSURANCE COVERAGE
================================================================================
  Provider                : AAR (80%, no cap)
  Calculated              : 50,000 x 80% = 40,000 UGX
  Insurance Coverage      : 40,000 UGX
  Patient Pays            = 50,000 - 40,000 = 10,000 UGX

================================================================================
   PATIENT BILL
================================================================================
  Patient                 : David Mugisha
  Service                 : Consultation
  Duration                : 1 hour
  Service Charge          : 50,000 UGX
  TOTAL BILL              : 50,000 UGX
  Insurance Provider      : AAR
  Insurance Coverage      : 40,000 UGX
  Patient Pays            = 10,000 UGX
================================================================================
Add another patient? (yes/no): no
```

TERMINAL — PART 6 OF 6 (Daily report)

```
================================================================================
   DAILY HOSPITAL REPORT
================================================================================
  Total Patients Served    : 5
  Total Revenue Collected  : 2,202,000 UGX
  Average Bill Per Patient : 440,400.00 UGX
  Total Insurance Coverage : 928,000 UGX
  Service Breakdown        : Consultation: 2 | Emergency: 0
                             Surgery: 2 | Admission: 1
  Most Common Service      : Consultation / Surgery (tie)
  With Insurance           : 4
  Without Insurance        : 1
  Most Common Provider     : AAR
================================================================================
```

Marking Scheme

Part	Marks	Criteria
A	6	All 8 prompts in order, validated, status checked against provider
B	6	All service rates; graduated hours/days and flat surgery bands
C	6	Insurance percentages with the NHIF and Jubilee caps
D	6	Lab and medication charges added before insurance
E	6	Bill with every label, one TOTAL BILL, patient pays
F	6	Report with revenue, 2-decimal average, ties, provider statistic
Total	**36**	

100% is only awarded if all parts are completed and every test case matches exactly. Master the tiered pricing and the insurance caps.